

ALY SADEK

MECHANICAL ENGINEER

951-517-9609 | Alyksadek@outlook.com | www.linkedin.com/in/alysadek

EDUCATION

Mechanical Engineering B.S. | California State University, Long Beach

August 2025 – Dec 2027

- Relevant Coursework: Deformable Bodies, Dynamics, Kinematics & Dynamics of Mechanisms, Manufacturing Processes, Project Cost-Benefit Analysis

TECHNICAL SKILLS

SolidWorks, AutoCAD, MATLAB, GD&T | CNC and Manual Mill & Lathe, 3D Printing | Microsoft 365 (Excel, Word, PowerPoint) | Experimental Design, Data Analysis, Technical Documentation, Project Coordination | Arabic

WORK EXPERIENCE

MacREU Undergraduate Research Internship | UC Riverside

June 2025 – August 2025

NSF-funded research on thin films and monolayer materials:

- Led project on direct lithium extraction from geothermal brines to evaluate lithium recovery efficiency.
- Investigated sodium-ion battery anodes, comparing hard carbon types for performance improvement.
- Utilized SEM, TEM, Raman, and XRD for material characterization.
- Collaborated with graduate researchers and Principal Investigators; refined methods, analyzed results, and presented findings at 2 technical research conferences (UCR Symposium & SCCUR).

Hot Air Balloon Ground Crew | CALIFORNIA DREAMIN', Temecula, CA

October 2023 – March 2025

- Operated as ground crew for commercial hot air balloon flights, managing equipment setup, launch, chase, and landing operations.
- Greeted passengers and delivered safety briefings and narration as part of the flight experience.
- Maintained mechanical systems including burners, fuel components, and balloon fabric.

STUDENT ORGANIZATION

Beach Launch Team | CSULB

Project Manager (Officer)

September 2026 – Present

- Elected by the general body to lead a 60+ member multidisciplinary student rocketry club; coordinate subsystem leads, project milestones, and team-wide execution.
- Run lead meetings, track deliverables, assign action, and coordinate technical and organizational priorities.
- Lead general-body presentations, recruiting, outreach, and team visibility across campus and external channels.
- Manage the club's administrative operations by securing grants, tracking funds, maintaining university regulations and safety compliance, coordinating with faculty advisors

GNC/Recovery Co-lead

September 2025 – September 2026

- Launched liquid-fuel Theseus rocket at Friends of Amateur Rocketry after 3-day desert field deployment; main chute failed, but oversized safety margins on dual drogue alone recovered ~85% of the rocket intact.

PROJECTS

- Rocket Recovery System Redesign, Petroleum System Research Analysis, MATLAB Monte Carlo Reliability Analysis